Recall Bias Toward Highly-Cited and English-Language Papers in AI Literature Curation

Abstract

Artificial intelligence (AI) research is increasingly curated through search engines, scholarly recommender systems, citation graphs, automated screening tools, and large language models (LLMs). These systems promise to reduce the cost of navigating a rapidly expanding literature, but they may also alter which papers are visible and subsequently incorporated into reviews, benchmarks, surveys, and research agendas. This paper develops the concept of **recall bias toward highly-cited and English-language papers**: a systematic disparity in the probability that an otherwise relevant paper is retrieved, surfaced, screened, or retained as a function of citation prominence and publication language. The concept connects established literatures on citation bias, cumulative advantage, language bias, bibliographic database coverage, scholarly recommender-system bias, and automated literature screening.

The central argument is that citation prominence and English-language status can operate as *upstream exposure variables* that influence literature discovery even when they are not explicit inclusion criteria. Highly cited articles may receive disproportionate exposure through citation-based search, recommender systems, ranking features, and human seed selection. English-language papers may likewise receive greater exposure because international scholarly communication, indexing, metadata, and retrieval infrastructure remain strongly English-dominant. Existing scholarship demonstrates the plausibility of both mechanisms, but direct empirical evidence quantifying their interaction specifically in AI literature curation remains limited. Recent studies of automated citation searching and LLM-assisted screening further show that retrieval performance is task- and corpus-dependent, making recall auditing essential rather than assuming that high precision implies adequate coverage.

The paper proposes a framework for measuring group-conditional recall, identifies methodological challenges in constructing a representative AI literature benchmark, reviews current human-AI and citation-network approaches, and outlines future research directions including multilingual retrieval, popularity-blind re-ranking, citation-stratified evaluation, temporal audits, and transparent human-in-the-loop workflows. The paper concludes that efficient AI literature curation should optimize not only relevance and workload reduction but also *representativeness of the evidence space*.

Keywords: artificial intelligence; literature curation; information retrieval; recall bias; citation bias; popularity bias; language bias; scholarly recommender systems; large language models; systematic literature review; multilingual retrieval

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1. Introduction

The contemporary AI research ecosystem has created a literature-discovery problem of unusual scale. Researchers encounter papers through bibliographic databases, general scholarly search engines, citation indexes, recommendation systems, social platforms, preprint repositories, and increasingly through AI-based research assistants. The fundamental challenge is not merely retrieving *some* relevant documents, but identifying a sufficiently representative subset of the relevant literature for a particular research question. This distinction is critical because the downstream conclusions of a survey, systematic review, benchmark, position paper, or research program depend on which papers become visible in the first place.

The problem can be expressed in conventional information-retrieval terms. Let (R) be the set of relevant papers in an underlying literature universe and (S) the set retrieved by a curation procedure. Recall is

$$
Recall = \frac{|R \cap S|}{|R|}.
$$

However, aggregate recall can conceal systematic disparities. Suppose (G) denotes a paper characteristic, such as citation prominence or publication language. Then the relevant quantity becomes

$$
Recall_g=P(S=1\mid R=1,G=g).
$$

A **recall bias** exists when (Recall_g) differs substantially across groups even after controlling, as far as possible, for substantive relevance. The phenomenon considered here is therefore not ordinary low recall. It is *selectively low recall*: papers that are relevant to the same research question may have different probabilities of being discovered because they are less cited, less visible in dominant databases, or published in languages other than English.

This paper focuses on two characteristics that may be particularly consequential in AI literature curation: **citation prominence** and **English-language status**. The first is connected to the long-standing Matthew effect and to empirical evidence that scientific citations are not determined exclusively by scientific relevance or quality. Merton's formulation of cumulative advantage emphasizes the tendency for already recognized scientific contributions to accumulate additional recognition, while more recent empirical work continues to identify citation predictors unrelated to scientific merit. The second is associated with the historically consolidated position of English as the dominant language of international scholarly communication and with the uneven indexing of non-English research in major databases.

Importantly, the thesis advanced here is not that highly cited or English-language papers are intrinsically less valid. Nor does it imply that citation counts or English-language indexing should never be used. Highly cited papers can be genuinely influential, and English remains a practical lingua franca for global scholarship. The concern is different: **when popularity and language become correlated with exposure, retrieval systems may systematically narrow the visible literature before substantive relevance has been fully assessed.**

The issue has become more urgent with AI-assisted curation. Traditional search strategies require information specialists and researchers to formulate queries and screen results. Modern systems add another layer: semantic embeddings, learning-to-rank models, citation-graph traversal, recommender algorithms, active learning, and LLM-based classification. These techniques can reduce screening workload substantially. For example, a large empirical study of RobotAnalyst found that active prioritization reduced the number of references required to identify 95% of abstract-level inclusions in most of the evaluated review collections, while also highlighting the need for stopping criteria and safeguards against reviewer complacency. Recent comparisons of supervised machine learning and LLMs show that automated screening can achieve recall close to or above human-reviewer recall on some datasets, but performance varies considerably by corpus and model.

The central research question is therefore:

> **To what extent can AI-enabled literature-curation systems systematically under-retrieve relevant AI papers that are less cited and/or non-English, and what methodological safeguards can prevent this from distorting the research record?**

The paper addresses this question through a conceptual synthesis of research on citation bias, language bias, bibliographic indexing, scholarly recommender systems, citation searching, automated evidence retrieval, and AI-assisted screening. It argues that the most important research gap is not simply the existence of individual biases but their **interaction across the complete curation pipeline**.

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2. Background and Related Work

2.1. Citation bias and cumulative advantage

Scientific literature is not a neutral network in which every relevant paper has equal probability of being discovered. Citations themselves are socially and structurally generated. Merton's concept of the Matthew effect describes cumulative advantage in scientific recognition: contributions that have already acquired prominence can receive disproportionate additional recognition. Subsequent research has refined this concept and shown that citation counts reflect several processes, including intellectual appropriateness, prestige, networking, field structure, and visibility. Ma's analysis of the Matthew effect, for example, found that citation outcomes could not be reduced to a single mechanism and that controlling for appropriateness substantially reduced some estimates of prestige effects.

Citation bias can also operate through the *content* of cited research. A systematic review and meta-analysis of 52 empirical studies found that statistically significant results were cited approximately 1.6 times as often as nonsignificant results, while articles explicitly reporting support for their hypotheses were cited even more frequently. The authors concluded that citation bias can contribute to overrepresentation of positive findings and the formation of unfounded beliefs. This evidence matters for literature curation because citation counts are not simply passive measurements of scholarly relevance: they are partly the product of prior selection behavior.

A 2025 scoping review of citation-rate research reinforces this point. Across 165 studies and 54 distinct predictors, more than half of the quantitative analyses identified factors associated with increased citation rates that were not straightforward measures of scientific quality or relevance. The authors therefore caution against interpreting citation metrics as uncomplicated indicators of scholarly merit.

These findings do not establish that citation-based retrieval is biased in every AI search environment. They do establish a theoretically plausible feedback structure:

$$
Visibility_t \rightarrow Citations_t \rightarrow Visibility_{t+1}.
$$

Once such reinforcement exists, a ranking system that incorporates citations, citation-network centrality, or user engagement may reproduce historical inequalities rather than independently assessing relevance.

2.2. Language bias and English-language dominance

The second relevant mechanism is linguistic. English has become the principal shared language of international scientific communication, particularly in the natural sciences. Histories of scientific communication describe this as a relatively recent consolidation: science was historically multilingual, whereas modern international scientific interaction is unusually concentrated around a single dominant language.

Empirical work shows that this preference is not merely a matter of individual choice. Stockemer and Wigginton surveyed more than 800 authors and found that non-Anglophone researchers submitted, on average, approximately 60% of their journal manuscripts in English, with substantial variation by field and region. The authors identified perceived increases in reputation and visibility as important motivations for publishing in English.

Database infrastructure can amplify the resulting concentration. Reviews comparing Web of Science and Scopus have documented historical biases toward English-language sources and weaker coverage of regional and non-English literature, although both databases have taken steps to broaden regional coverage. Comparisons among citation databases further show that coverage differs meaningfully across systems: Google Scholar, for example, can capture substantial additional material, including non-journal and non-English citing sources that are absent from Web of Science or Scopus.

The consequences of language restriction are context-dependent. A systematic review of empirical studies found no consistent major difference in pooled treatment effects between English-restricted and language-inclusive meta-analyses in conventional medicine, but inclusion of non-English studies improved precision and the individual importance of such studies was difficult to predict. Other empirical analyses have found that non-English trials can have different effect distributions, demonstrating why blanket assumptions about their irrelevance are unsafe.

The appropriate conclusion is therefore not that every non-English publication is essential. Rather, **language cannot safely be treated as an unbiased proxy for relevance**.

2.3. Linguistic and content effects within AI itself

The AI field provides additional reason to investigate language- and content-mediated selection. Vincent-Lamarre and Larivière analyzed more than 12,000 AI-related manuscript submissions and found systematic associations between linguistic characteristics, topical vocabulary, citation patterns, and peer-review outcomes. Accepted manuscripts were more likely to use AI and machine-learning terminology and exhibited greater similarity in their referencing patterns. The authors also discuss possible language and content bias while emphasizing that their analyses are correlational.

Experimental work outside AI likewise suggests that linguistic form can influence scholarly evaluation. In a randomized study using otherwise identical scientific abstracts, Politzer-Ahles, Girolamo, and Ghali found preliminary evidence that abstracts written in less “native-like” academic English could receive lower quality ratings, although the authors stress that some results were statistically inconclusive. These findings are relevant because retrieval models are trained on corpora produced through earlier stages of scholarly selection; any language-related inequalities entering at publication or peer review can subsequently propagate into data used by automated systems.

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3. From Citation and Language Bias to Recall Bias in AI Literature Curation

3.1. A pipeline perspective

AI literature curation is better understood as a multi-stage pipeline than as a single search query:

[
\text{Indexing}
\rightarrow
\text{Candidate generation}
\rightarrow
\text{Ranking}
\rightarrow
\text{Screening}
\rightarrow
\text{Citation chaining}
\rightarrow
\text{Retention}
\rightarrow
\text{Synthesis}.
]

Bias can enter at every stage.

At the **indexing stage**, non-English journals and regional venues can be underrepresented in major databases. At **candidate generation**, search terms and metadata fields privilege documents whose concepts are expressed in the query language. At **ranking**, citation counts, venue prestige, prior user interactions, or learned relevance models can increase exposure for already prominent papers. At **screening**, active-learning systems learn from early human decisions and may therefore amplify the initial composition of the labeled sample. Finally, citation chaining creates path dependence: papers discovered first can become seeds for discovering additional papers.

Consequently, a literature-curation system can achieve excellent precision on the documents it surfaces while still exhibiting poor *population recall*.

3.2. Scholarly recommender systems

Scholarly recommender systems directly demonstrate the relevance of this concern. Wang and Blei's influential collaborative topic-modeling framework was developed precisely because the increasing volume of online scientific articles made relevant-paper discovery difficult; notably, the authors explicitly described citation-following as effective but biased toward heavily cited papers.

More recent scholarship has identified numerous forms of bias in scholarly recommender systems. Färber, Coutinho, and Yuan classify biases according to whether they arise from human behavior or system design and identify popularity bias, selection bias, conformity effects, position effects, cold-start problems, and Matthew effects as particularly relevant to academic recommendation. They emphasize that scholarly systems differ from ordinary recommender systems because scientific documents carry citation structures, communities are fragmented by field, and the academic reward system itself contains cumulative-advantage mechanisms.

A 2024 survey of popularity bias in recommender systems similarly concludes that recommendation algorithms can systematically overexpose popular items while underexposing long-tail items and that mitigation research remains dominated by offline computational experiments. The scholarly domain is therefore a special case of a more general recommender-system problem: popularity can become a feedback signal that determines future exposure.

3.3. Citation searching and path dependence

Citation searching is not inherently undesirable. Indeed, methodological guidance recognizes backward and forward citation searching as valuable supplements to keyword searching, particularly for difficult-to-search topics. The TARCiS statement recommends citation searching when relevant evidence may be difficult to capture through conventional queries and stresses systematic reporting of the process.

The problem arises when citation-based expansion begins from a **nonrepresentative seed set**. If the first seeds are selected because they are highly cited, highly visible, or already familiar to reviewers, then the citation graph becomes a biased sampling mechanism. A paper that is weakly connected to the seed network can remain invisible even if it is highly relevant. This is particularly consequential across disciplinary boundaries, where different research communities may use different terminology and cite different foundational work.

Empirical work confirms that citation searching changes retrieval composition. In one case study, citation tracking across Google Scholar, Scopus, Web of Science, and another database required multiple resources to recover all identified citations, demonstrating that citation-index coverage itself can affect retrieval. A 2023 study of qualitative systematic reviews similarly found that supplementary citation and related-document strategies recovered many publications that traditional database searching had missed.

Thus citation chaining should be treated as a *recall-enhancement technique*, not automatically as a representative sampling strategy.

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4. Current Approaches and Developments

4.1. Active learning and machine-assisted screening

Technology-assisted systematic review tools seek to reduce the human burden of screening. RobotAnalyst, for instance, uses text mining, topic modeling, relevancy classification, and active prioritization to order references for human review. Its evaluation across 22 reference collections demonstrated substantial workload reduction in many collections, but also highlighted the importance of stopping criteria and human vigilance.

ASReview and related active-learning systems similarly treat screening as an interactive process in which model predictions are updated from reviewer decisions. The technical attraction is clear: rather than examining records at random, researchers can review those estimated to be most promising first.

However, active learning creates an important tension for recall-bias research. If the initial labeled examples contain mostly English and highly cited papers, the system may rapidly learn that these characteristics correlate with inclusion—not because they cause substantive relevance, but because they are overrepresented in the early training data. The system can then prioritize similar papers and push other groups downward in the queue.

This is a form of **selection-induced model bias**. It differs from conventional popularity bias because the popularity signal need not be explicitly encoded in the algorithm; it can enter through the distribution of the training labels.

4.2. Automated citation searching

Recent work has begun to automate citation-network exploration directly. Rajit and colleagues simulated automated citation searching using OpenAlex and Semantic Scholar across 27 systematic reviews. Automated citation searching produced higher precision and F1 scores than the reference standard in their evaluation, but significantly lower recall and F3 scores, leading the authors to recommend citation automation primarily as a supplementary strategy when recall is especially important.

This result is especially relevant to the present argument. A system can appear efficient because it sharply reduces the number of records that humans must inspect, yet still fail to recover a meaningful minority of relevant papers. Efficiency metrics therefore cannot replace recall evaluation.

The recent development of automated multi-level citation-search tools, including systems designed to expand backward and forward citation networks with relevance ranking, suggests a promising trajectory. Such systems can improve completeness, but their evaluation must explicitly test whether performance differs by citation count, venue, language, geographic origin, age, and disciplinary community. Otherwise, high overall sensitivity can conceal low group-conditional recall.

4.3. Large language models and semantic retrieval

LLMs and embedding-based retrieval offer another possible route around keyword and metadata limitations. Semantic retrieval can identify papers that use conceptually related but lexically different terminology, an important advantage in fast-moving interdisciplinary AI fields.

Evidence published in 2026 indicates that GenAI is increasingly useful across several systematic-review tasks. A rapid review of 115 studies found the strongest evidence for title/abstract screening and data extraction, with selected high-quality evaluations reporting sensitivity of at least 90% and substantial workload reductions in calibrated human-in-the-loop workflows. At the same time, performance for search strategy development, full-text screening, risk-of-bias assessment, and qualitative synthesis was more variable, and fully autonomous end-to-end review generation was considered unreliable.

A 2026 comparative study of supervised machine-learning models and Llama-3.1 for title/abstract screening provides a similar message. Recall differed substantially across datasets; the LLM was comparatively sensitive, while several supervised methods achieved better balances between recall and specificity.

These findings imply that the appropriate evaluation question is not simply “Does the model work?” but rather:

$$
\text{Does the model work equally well across relevant literature strata?}
$$

A multilingual model may still produce lower recall for documents with unusual metadata. A powerful embedding model may still learn popularity correlations from training data. A generative system may summarize only sources retrieved by a biased upstream search.

4.4. Popularity-aware versus popularity-blind ranking

One possible mitigation is to separate **relevance ranking** from **exposure ranking**. Citation count could be withheld from the initial retrieval stage, then reintroduced only as a secondary exploratory signal.

For example, a two-stage system could compute

$$
Score(p)=\alpha \cdot SemanticRelevance(p)+
\beta \cdot CitationDiversity(p)+
\gamma \cdot LanguageCoverage(p),
$$

where (β) and (γ) are designed to reward *coverage* rather than raw popularity.

A related strategy is **citation-stratified sampling**: divide the candidate pool into citation bands and ensure that each band contributes a sufficient number of candidates for human assessment. An analogous language-stratified design can ensure systematic testing of non-English records and papers whose metadata are multilingual or locally indexed.

Such mechanisms should not be interpreted as artificially promoting low-quality papers. Their purpose is to create **opportunities for evaluation** independent of historical visibility.

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5. Research Challenges and Limitations

5.1. No universally known denominator

Recall requires an estimate of the relevant universe (R), but in real-world AI literature no complete gold-standard corpus exists. This is a fundamental methodological problem. If a paper was never retrieved, there is often no independent label proving that it should have been included.

Possible approximations include expert-built benchmark corpora, comprehensive review collections, venue-defined corpora, citation-network exhaustions, or union sets produced by multiple independent retrieval methods. Each has limitations. A benchmark constructed from existing reviews may inherit the same popularity and language biases under investigation.

5.2. Citation count is endogenous

Citation count is correlated with visibility, age, field, journal prestige, author reputation, social networks, and research topic. Therefore, dividing papers into “highly cited” and “poorly cited” groups can conflate several causal mechanisms.

A paper may be highly cited because it is genuinely foundational. Another may be highly cited because it introduced a benchmark widely used by the field. Yet another may be highly cited because it reported a striking positive result. Citation-stratified analysis must therefore include controls for publication age, venue, subfield, document type, and topic.

5.3. Language and quality are difficult to disentangle

Non-English literature is not homogeneous, just as English literature is not homogeneous. Some papers may be locally important but invisible internationally; others may be less relevant to the research question. Translation quality, metadata availability, indexing practices, and field-specific terminology further complicate analysis.

The evidence from systematic-review methodology suggests that language restrictions do not always materially alter conclusions, but the value of non-English evidence varies by topic and cannot be known a priori. The correct methodological response is therefore empirical measurement rather than assuming either universal importance or universal dispensability.

5.4. Model improvements may worsen representativeness

A system optimized exclusively for precision, F1, user satisfaction, or screening-time reduction can become more selective as it becomes more efficient. This produces an apparent paradox:

$$
Efficiency \uparrow
\quad \not\Rightarrow \quad
Representativeness \uparrow
$$

and in some circumstances,

$$
Precision \uparrow
\quad \text{while} \quad
Recall_{long-tail} \downarrow.
$$

This is particularly important because current AI-assisted systematic-review research often evaluates models within predefined benchmark datasets. The benchmark itself may not reveal systematic failure on underindexed or non-English literature.

5.5. Temporal bias

Citation count is inherently time-dependent. Older papers have had longer to accumulate citations, while new AI papers may be highly relevant but have low citation counts simply because they are recent. A citation-heavy ranking strategy can therefore systematically disadvantage new work.

Conversely, a system trained on historical literature can disproportionately reproduce established paradigms. In AI, where paradigms can shift rapidly, the distinction between *historical prominence* and *current relevance* is especially important.

5.6. Interactions between stages

The most serious limitation of current research is that biases are usually measured in isolation. Database coverage studies emphasize language; bibliometrics emphasize citation counts; recommender-system research emphasizes popularity; systematic-review research emphasizes recall; and LLM studies often emphasize classification accuracy.

But the actual process is multiplicative. A non-English paper may be underindexed, have fewer international citations, be less likely to become a seed document, and then be ranked lower by an AI recommender. The combined effect could be substantially larger than any single-stage effect.

6. Research Gaps and Future Directions

6.1. Establish a benchmark for group-conditional recall

The field needs a large, versioned benchmark containing AI papers across:

1. citation percentiles;
2. publication languages;
3. geographic regions;
4. venues and publishers;
5. subfields;
6. document age;
7. document type;
8. open-access versus paywalled status.

The benchmark should have independent expert relevance labels and multiple retrieval pathways. Crucially, the benchmark should report both aggregate recall and conditional recall:

$$
R_{citation,q}, \qquad R_{language,l}.
$$

6.2. Measure interactions rather than isolated effects

Future studies should estimate models of the form

$$
P(S=1)=\sigma(
\beta_0+
\beta_1 Citation+
\beta_2 Language+
\beta_3 Age+
\beta_4 Field+
\beta_5(Citation\times Language)
+\cdots).
$$

The interaction term is particularly important. A non-English, low-citation paper may face a much greater retrieval disadvantage than either characteristic alone would predict.

6.3. Separate retrieval from ranking

AI literature systems should report at least three metrics separately:

- **candidate-generation recall**: was the paper retrieved at all?
- **ranking exposure**: how highly was it surfaced?
- **screening recall**: was it ultimately retained?

This decomposition would reveal whether bias originates in indexing, retrieval, ranking, or human-AI screening.

6.4. Develop multilingual and cross-lingual retrieval

Cross-lingual retrieval should become a first-class requirement for international evidence synthesis. Rather than simply translating English queries, future systems should support multilingual embeddings, translated metadata, local terminology, transliteration, and language-specific citation networks.

Evaluation should measure whether translation-based expansion recovers substantively relevant papers without introducing excessive noise. Importantly, non-English papers should not have to be translated into fluent English before they become visible to the ranking system.

6.5. Introduce popularity-blind auditing

A practical audit can rerun the same curation task under two conditions:

$$A_1=\text{system with citation information},$$

$$A_2=\text{system without citation information}.$$

The change in group-conditional recall provides a direct empirical estimate of how much retrieval depends on popularity signals. A similar experiment can manipulate language metadata or compare monolingual and multilingual retrieval.

6.6. Use counterfactual evaluation

Counterfactual tests can create matched paper pairs that are substantively similar while varying citation count, language, or venue prominence. If ranking probabilities change systematically after such manipulation, the result provides stronger evidence of causal sensitivity.

For example, a paper's citation score could be replaced with a normalized value corresponding to another age- and field-matched paper while holding semantic content constant. Such interventions would help distinguish genuine relevance effects from exposure effects.

6.7. Preserve human oversight, but diversify human seeds

Human-in-the-loop workflows remain appropriate. Current evidence favors calibrated hybrid approaches rather than fully autonomous systematic reviews. However, human oversight is not automatically unbiased. Reviewers tend to select familiar seminal papers as seeds, and this choice can initiate popularity-driven citation cascades.

A better design would require seeds from multiple sources: keyword retrieval, subject experts, recent papers, low-citation strata, multilingual sources, and independent databases.

6.8. Report retrieval provenance

Every AI-assisted literature review should ideally disclose:

- databases and repositories searched;
- date of each search;
- languages included;
- citation and popularity signals used;
- ranking or filtering criteria;
- model version;
- stopping criteria;
- numbers screened and retained;
- recall estimates on known benchmark papers;
- records recovered uniquely by supplementary strategies.

The TARCiS guidance provides an important methodological precedent by emphasizing standardized terminology and transparent reporting for citation searching.

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7. Conclusion

AI literature curation offers major benefits for a research community facing unprecedented information growth. Semantic retrieval, recommender systems, citation graphs, active learning, and LLMs can reduce search and screening costs that would otherwise be prohibitive. Current evidence supports their usefulness, especially when deployed as human-in-the-loop tools rather than unrestricted autonomous reviewers.

At the same time, efficient retrieval is not synonymous with representative retrieval. The scientific literature already exhibits cumulative citation advantage, non-random citation behavior, English-language dominance, and uneven database coverage. Scholarly recommender-system research further shows that popularity and exposure can become mutually reinforcing.

This paper has framed the resulting concern as **recall bias toward highly cited and English-language papers**. The concept should be understood as a testable hypothesis and an evaluation framework rather than as an already-established universal empirical law. Existing evidence strongly supports the plausibility of the underlying mechanisms, but direct, large-scale measurement of their combined effect in AI literature curation remains an important research gap.

The central methodological implication is straightforward: future AI curation systems should be evaluated not only by overall precision, F1, or workload reduction, but by **who is missing from the retrieved literature**. A system that repeatedly returns the same influential English-language papers may feel exceptionally useful because those papers are easy to recognize and highly relevant to the dominant research narrative. Yet the most consequential omissions may occur precisely among papers with weaker citation histories, less international visibility, alternative terminology, or non-English publication venues.

For AI research to remain epistemically pluralistic and internationally representative, retrieval systems must therefore treat *coverage as a scientific objective*. Popularity should be measured, not mistaken for relevance; language should be accommodated, not treated as a proxy for quality; and automation should be audited for the distribution of what it fails to retrieve. The next generation of scholarly AI should optimize not merely for finding the papers most likely to matter according to the existing literature, but for discovering evidence that the existing literature may have systematically overlooked.

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